Physics Questions

Kinematics and Vectors

Questions generated below. Multiple-choice answers in RED; incorrect answers in BLUE.

Q1. A survey drone is located at the origin of an xy-coordinate system. At a certain instant, a weather balloon is at position $\vec{r} = -8\hat{i} + 15\hat{j}$, where distances are measured in kilometers. Which of the following statements is correct?

A. The distance between the balloon and the drone is 23 km.
This comes from adding the component magnitudes, $8 + 15 = 23$, instead of finding the vector magnitude. Distance from the origin in two dimensions must be computed with the Pythagorean relation, not simple addition.

B. The distance between the balloon and the drone is 17 km.
The distance from the origin is the magnitude of the position vector. Using $|\vec{r}| = \sqrt{(-8)^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17$ km, the correct statement is that the balloon is 17 km away. The negative sign on the x-component affects direction, not the final distance.

C. The distance between the balloon and the drone is 15 km.
This mistake uses only the y-component and ignores the horizontal displacement. In two-dimensional motion, both components contribute to the distance from the origin.

D. The distance between the balloon and the drone is 8 km.
This mistake uses only the magnitude of the x-component. The full distance must include both perpendicular components of the position vector.

Answer: *B*

Q2. The position of a particle moving along the x-axis is given by $\vec{s}(t) = \left[(0.90\,\text{m/s}^3)t^3 + (4.2\,\text{m/s}^2)t^2\right]\hat{i}$. Which of the following statements is correct?

A. This motion has constant acceleration.
This error comes from assuming any polynomial position function leads to constant acceleration. Because the position contains a t^3 term, the second derivative still depends on t, so the acceleration is not constant.

B. The magnitude of the acceleration at $t = 0$ is $4.2\,\text{m/s}^2$.
This mistake reads the coefficient of the t^2 term directly as the acceleration. The acceleration must be found by differentiating the position function twice.

C. The magnitude of the acceleration at $t = 0$ is $8.4\,\text{m/s}^2$.
Differentiate the position to get velocity: $\vec{v}(t) = \dfrac{d\vec{s}}{dt} = \left[(2.7)t^2 + 8.4t\right]\hat{i}$. Differentiating again gives acceleration, $\vec{a}(t) = \dfrac{d\vec{v}}{dt} = [5.4t + 8.4]\,\hat{i}$, so at $t = 0$ the magnitude is $8.4\,\text{m/s}^2$. The motion therefore has time-dependent acceleration, but its initial acceleration is nonzero.

D. The magnitude of the acceleration is $5.4\,\text{m/s}^2$.
This comes from confusing the coefficient of the linear term in $a(t)$ with the actual acceleration value. Since $a(t) = 5.4t + 8.4$, the acceleration depends on time and is not always $5.4\,\text{m/s}^2$.

Answer: *C*

Q3. A robot moves for 4.0 s from position $\vec{r}_A = 1\hat{i} - 7\hat{j}$ to position $\vec{r}_B = -5\hat{i} + 1\hat{j}$, where positions are measured in meters. Which of the following statements is correct?

A. The acceleration at $t = 0$ must be 0.63 m/s^2.
This is a misconception because acceleration cannot be uniquely determined from only initial position, final position, and elapsed time. Many different motions with different accelerations could connect the same two points in 4.0 s.

B. None of the above is correct.
From the given data, the displacement is $\Delta\vec{r} = \vec{r}_B - \vec{r}_A = (-6)\hat{i} + 8\hat{j}$, whose magnitude is $|\Delta\vec{r}| = \sqrt{(-6)^2 + 8^2} = 10$ m. However, this 10 m is the displacement magnitude, not necessarily the distance traveled, and the information provided is insufficient to determine the instantaneous speed or acceleration at $t = 0$. Therefore none of the other statements must be true.

C. The distance traveled by the robot is 10 m.
This confuses displacement magnitude with path length. The straight-line displacement is 10 m, but the actual distance traveled depends on the path and cannot be known from the endpoints alone.

D. The speed at $t = 0$ must be 2.5 m/s.
This comes from dividing the displacement magnitude by the total time, which gives only a rough average based on endpoint information. Instantaneous speed at the initial moment cannot be determined without more information about the motion.

Answer: *B*